

A Review of Current Tidal Energy Research

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Abstract—This literature review examines the current state of tidal and wave energy technologies, evaluating their technical viability, economic feasibility, and environmental implications. Drawing on six key papers—including Chen (2023)’s foundational comparison of marine energy forms and five other contemporary studies—this review synthesizes findings across predictive modeling, prototype performance, commercialization trajectories, and policy frameworks. Tidal energy, due to its gravitationally driven predictability, emerges as particularly promising for grid stability and hybrid renewable systems, especially in coastal and island contexts. Despite this potential, the sector faces persistent challenges: high capital expenditure, environmental assessment complexity, lack of standardization, and limited field-scale testing data. Innovations in turbine design, multi-use offshore platforms, and site-specific modeling signal pathways to improved efficiency and broader adoption. However, unresolved questions remain regarding long-term ecological impacts, climate-resilient resource modeling, and economically viable deployment at scale. The review concludes that while tidal and wave energy systems are still technologically immature compared to solar or wind, strategic investment in modular design, policy alignment, and environmental integration may unlock their role as stable, low-carbon contributors to future energy systems.

I. INTRODUCTION

This comprehensive review covers all aspects of the current research landscape of tidal energy and tidal energy converters. It begins with a discussion of the most current contributions made to the field, as related to the predictability of tidal energy, the economics and cost of this technology, the most recent technological developments, the environmental and geographical concerns, the integration of tidal energy into the existing power grid, the governmental policy surrounding the technology, and the problem of commercialization that tidal energy converters currently face. The review further discusses the limitations and gaps within the current research, including a lack of tidal energy model standardization or deployment and a lack of discussion around the managing of the intermittent nature of tidal energy or the corrosion and maintenance costs that may arise. The review ends with a discussion of the current unanswered question in the field and possible extensions for further research.

Six separate research papers were used to inform all sections of the review. The papers were selected to cover a broad range of topics and viewpoints on tidal energy and tidal energy research. “Wave Energy: A Pacific Perspective,” by Paasch et al. provides broad information on current tidal energy designs as well as the permitting process associated with them. “A Large-Scale Review of Wave and Tidal Energy Research Over the Last 20 Years,” by Khojasteh et al. is a comprehensive study of all research in this field in recent history. This paper provides a practical view of current tidal designs, largely from an economic standpoint. Also coming from an economic standpoint, “Financial and Economic Assessment

of Tidal Stream Energy – A Case Study,” by Klaus, is a detailed look into the economic feasibility of tidal energy converters as a whole. “Wave and Tidal Energy: A Patent Landscape Study,” by Pazhouhan et al., provides a viewpoint of this technology from a judicial perspective, looking into the governmental and bureaucratic processes required of this technology. “Marine Energy: Comparison of Tidal and Wave Energy,” by Chen provides a clear outline of the definitions, benefits, and drawbacks to tidal energy, especially as compared to wave energy. Finally, “Evaluation of Tidal Energy Potential Using a Two-Way Tidal Energy Model,” by Rusvan et al., explores a physical example of model testing of new tidal energy designs.

Tidal energy is an optimistic, yet largely untapped energy source. Using the six papers listed prior, this review covers aspects of the latest research related to tidal energy, painting a picture of why tidal energy has not been harnessed to its full potential and what the future may look like for this technology.

II. TECHNICAL CONTRIBUTIONS

A. Tidal Predictability

Tidal power’s most distinctive engineering asset is its astronomical determinism. Gravitational interactions among the Earth-Moon-Sun system create semidiurnal and diurnal cycles that can be predicted, with minute-scale accuracy, decades in advance, a capability no wind or irradiance-driven resource can match. Grid operators can therefore model ramp rates, slack-tide windows, and maintenance outages with exceptional confidence, a feature that is especially attractive for coastal microgrids and Small-Island Developing States (SIDS) where fuel logistics and resilience premiums dominate the cost stack. Empirical evidence supports this deterministic promise. Prototype arrays at semidiurnal sites routinely deliver up to 20 h of generation per day, and laboratory validation shows that calibrated flow-height models maintain $\pm 3\%$ error margins when extrapolated to full-scale rotors [1]. Rusvan et al. extend these findings with a two-way barrage experiment that links basin geometry to power output, demonstrating how fluid-structure interactions can be forecast once bathymetry is fixed [1]. Such high-fidelity forecasting enables tighter sizing of batteries or diesel back-ups compared with equally remote solar-PV systems. Predictability, however, is not continuity. Slack tides interrupt generation for 15-90 minutes twice per cycle, and seasonal spring-neap variations can shrink daily production by up to four hours [1]. Hybridization strategies therefore pair tidal stream energy (TSE) with wave converters or solar PV to fill these cyclic troughs, transforming the resource from a periodic to a pseudo-baseload contributor. Early simulations of tidal-PV-battery microgrids confirm lower curtailment and smaller storage requirements than PV-only

baselines, but field data remain sparse because most hybrid pilots are still at the demonstrator stage. A second nuance is spatial predictability versus spatial granularity. Large-scale resource atlases accurately forecast regional tidal envelopes, yet high-resolution flow datasets are still lacking for many equatorial archipelagos and estuaries, constraining bankable site selection and turbine optimization workflows. Advanced resource characterization campaigns that fuse Acoustic Doppler Current Profiler (ADCP) transects, satellite altimetry, and CFD wake modeling are therefore an urgent research priority if tidal projects are to move beyond coastlines of countries a part of the Organization for Economic Co-operation and Development (OECD). The deterministic nature of tides dramatically simplifies scheduling and hybrid-system dispatch, but these advantages only translate into investor confidence when lifetime costs can compete with incumbent diesel or wind. The next section (“Economic Burden”) examines how high CAPEX, proprietary performance data, and limited supply-chain learning curves currently offset predictability’s economic upside and what techno-economic levers can narrow that gap.

B. Economic Burden

The deterministic nature of tides translates into operational certainty, yet certainty alone does not guarantee affordability. Today’s tidal-stream projects still struggle to convert their scheduling advantage into competitive dollars-per-kilowatt-hour because the cost stack—capital, financing, and maintenance—remains sharply higher than for solar-PV, onshore wind, or even offshore wind. This section dissects the economic headwinds, grounded in recent techno-economic models and field evidence, and sets the stage for the technological and policy levers discussed later.

1) *Capital Cost Anatomy*: A single commercial-scale tidal array demands heavy civil works (foundations or anchors), bespoke turbine fabrication, subsea cabling, grid interconnection, and specialized installation vessels. Because production volumes are low, there are few opportunities for supplier bargaining or modular manufacturing. Klaus estimates that turbine and support-structure CAPEX account for roughly 60% of total installed cost, with export cables and grid upgrades adding another 15%; onshore renewables typically invert that ratio, devoting less than a third of CAPEX to foundations and balance-of-plant [2]. Site heterogeneity compounds the problem: every strait or estuary has unique bathymetry and sediment dynamics, limiting design replication and denying developers the serial-production learning curves that drove wind-turbine prices down during the 2010s.

2) *Levelized-Cost Benchmarks*: The combined effect of high CAPEX and undeveloped supply chains keeps the levelized cost of electricity (LCOE) for tidal projects in the \$0.35–\$0.48 kWh band under present-day assumptions [2]. Broader literature surveys place early-stage wave-and-tidal ventures even higher—\$200–\$400 MWh (\$0.20–\$0.40 kWh)—while mature offshore wind averages \$40–\$60 MWh and utility-scale solar ranges from \$20–\$50 MWh. These disparities effectively price tidal out of merchant-market participation except in niche contexts—remote islands, Arctic bases,

or defense installations—where diesel alternatives exceed \$0.50 kWh once fuel logistics are included [2]. Klaus’s hybrid-microgrid simulations illuminate how predictability tempers cost: when tidal generation (20 h/day) is co-located with PV and battery storage, the dispatchable hours rise and storage oversizing falls, nudging hybrid LCOE below equivalent diesel baselines even at current turbine prices [2]. Yet the model’s sensitivity analysis shows that a 30% CAPEX reduction or 10-year debt tenor extension is necessary to challenge wind-solar hybrids on continental grids. In other words, technical synergy helps but cannot fully offset capital intensity.

3) *Financing Friction and Risk Premiums*: Because so few tidal plants have achieved multi-year service histories, investors demand steep risk premiums. Khojasteh et al. highlight the absence of standardized performance benchmarks and transparent OPEX data, which forces lenders to rely on conservative failure-rate assumptions and pushes weighted-average cost of capital (WACC) well above that of solar or wind projects [3]. The result is a vicious circle: high WACC inflates LCOE, which in turn limits deployment volume and starves the sector of the operational datasets that would de-risk future projects. Policy economists therefore argue for social discount rates (SDRs) and explicit carbon-externality pricing to correct market signals. Lower SDRs recognize the long-term public value of carbon-free baseload power, while carbon taxes or emissions-trading revenues help level the playing field against fossil peakers. Early adopters like the UK (Contracts for Difference) and Canada (tidal-specific innovation funds) illustrate how stabilized revenue streams can crowd in private capital, yet most jurisdictions still lack analogous instruments.

4) *Supply Chain Immaturity and Learning Curves*: Economies of scale remain elusive. Global installed tidal capacity sat at approximately 32 MW in 2020—roughly three commercial wind turbines’ worth of nameplate power. Such a small fleet produces neither volume discounts nor shared O&M ecosystems. Spare-parts logistics, diver operations, and cable-laying contracts are all costed as one-off bespoke services rather than commoditized packages. Modularization—a recurring theme across engineering and policy papers—offers a partial remedy. Quick-release moorings, surface-retractable nacelles, and floating platforms aim to reduce installation vessel time and enable maintenance in sheltered waters, thereby lowering OPEX and increasing turbine availability. Evidence from offshore-wind monopile standardization suggests that even modest component homogeneity can unlock 15 – 25% cost reductions over two to three manufacturing generations [3].

5) *Data Transparency and the Cost of Proprietary Knowledge*: Performance data, especially on blade fouling, gearbox fatigue, and power-curve degradation, are often treated as commercially confidential by prototype operators. Pazhouhan et al. warn that this secrecy throttles cumulative learning and slows convergence toward best-practice designs, extending the valley-of-death period between R&D and bankable plant sizes [4]. Open-access databases—analogue to the Global Wind Atlas—are therefore a strategic priority. Coupled with IEC 62600 certification protocols, such transparency would shorten due-diligence cycles, cut feasibility-study costs, and reduce

financier uncertainty.

6) *Forward Trajectory and Link to Technology Maturation*: Tidal energy's predictability offers quantifiable system-planning benefits, but present-day economics remain hamstrung by capital intensity, high financing costs, and a fragmented supply chain. Closing the gap demands both engineering breakthroughs—to trim CAPEX/OPEX—and financial-policy innovations—to shrink WACC and internalize climate externalities. The next section, Technology Maturation, examines how blade materials, power-electronics smart control, and modular array designs are evolving precisely to attack these economic pain points and tip the LCOE curve downward.

C. Recent Technological Developments

Technological innovation in tidal energy converters (TECs) is accelerating, as evidenced by both academic analysis and patent trends. Khojasteh et al. classify tidal turbine development into four major clusters: device design and hydrodynamic optimization, power electronics and control, array configuration, and hybrid systems [3]. These themes are corroborated by Pazhouhan et al., who use topic modeling to extract seven major innovation clusters from the global patent landscape, including improvements in vertical axis turbines, oscillating hydrofoils, and biofouling-resistant surfaces [4].

1) *Device Design and Hydrodynamic Optimization*: Device design and hydrodynamic optimization constitute the foundational pillar of tidal turbine development. The majority of commercially deployed tidal energy converters (TECs) to date are horizontal-axis turbines (HATs), which function analogously to underwater wind turbines and are favored for their high energy conversion efficiency in unidirectional, high-velocity flows. However, these systems often require active yaw control or bidirectional blade systems to operate effectively in reversing tidal currents, which increases mechanical and control complexity. Alternatives such as Darrieus-style cross-flow turbines have been investigated for their ability to handle multi-directional flow without active orientation systems, though their energy efficiency generally remains lower than HATs. Other innovations include oscillating hydrofoil systems, which convert lift forces into rotational or oscillatory motion, and tidal kites, which magnify water flow speed over a small turbine by dynamically sweeping through the current. The central focus of this cluster is the precise engineering of blade geometry, rotor dynamics, and structural materials to optimize the lift-to-drag ratio and minimize cavitation and fatigue loads [3]. Computational fluid dynamics (CFD) and physical model testing are frequently used to refine these designs. Despite considerable progress, device-level hydrodynamic optimization remains an ongoing area of research due to the diverse and often site-specific flow regimes found in coastal and estuarine environments.

2) *Power Electronics and Control Systems*: Power electronics and control systems represent a critical innovation cluster aimed at enhancing the operational reliability and grid compatibility of tidal energy devices. Tidal turbines operate in dynamic aquatic environments where flow velocity and turbulence levels vary significantly throughout the

day. To maximize power output under such variability, modern systems employ advanced control strategies that adjust torque, rotational speed, and blade pitch in real-time. These capabilities rely on robust sensor integration and feedback mechanisms, typically embedded within supervisory control and data acquisition (SCADA) frameworks [3]. Additionally, power electronic interfaces—such as inverters, rectifiers, and voltage regulators—are required to convert raw turbine output into stable, grid-compliant electricity. In microgrid contexts or weak-grid conditions, these systems must also manage reactive power, harmonic distortion, and islanding scenarios. Research in this cluster has also focused on fault-tolerant control schemes and automated shutdown protocols to prevent damage during extreme flow events. While the development of intelligent control and power conditioning systems has made substantial strides, challenges persist around system robustness, especially in corrosive marine environments where electronics are difficult to access and maintain.

3) *Array Configuration*: The array configuration cluster deals with the design, placement, and hydrodynamic interaction of multiple tidal turbines deployed within the same installation site. Similar to wind farms, tidal energy arrays must be strategically arranged to minimize wake effects, which can cause turbulence and reduce downstream turbine efficiency. However, the denser medium of water and the bidirectional nature of tidal flow introduce unique constraints on spacing and alignment. Researchers use CFD simulations, vortex modeling, and Blade Element Momentum (BEM) theory to optimize the layout of arrays in order to balance power extraction with environmental preservation and maintenance accessibility. Innovative layout strategies, including staggered and horseshoe-shaped configurations, are being tested to enhance energy capture while minimizing destructive interference. Furthermore, the structural design of moorings, anchors, and support platforms plays a critical role in both performance and cost. Recent work has also emphasized modularity, advocating for standardized, interchangeable support structures that simplify retrieval, inspection, and redeployment of turbines. The complexity of optimizing array configuration increases with the scale of the installation and the variability of the local bathymetry and tidal regime [3].

4) *Hybrid Systems*: Hybrid systems represent a forward-looking cluster focused on integrating tidal energy with other renewable sources, energy storage, and end-use applications. Due to the intermittent yet predictable nature of tidal flows, tidal stream energy (TSE) is ideally suited for integration with variable renewables such as solar photovoltaic (PV) and wind power. Studies demonstrate that combining TSE with solar and battery energy storage systems (BESS) can yield more stable and dispatchable power profiles, especially in remote coastal microgrids or Small Island Developing States (SIDS) that currently depend on diesel generation. Furthermore, multi-modal offshore energy platforms are being explored where tidal, wave, and wind energy converters share infrastructure, such as cabling and maintenance vessels, to reduce capital and operational expenditures [3]. Beyond energy generation, hybrid systems may also support applications such as seawater desalination, aquaculture, and coastal defense. These

systems require multi-layered control architectures capable of managing diverse generation profiles, storage dynamics, and load priorities. Although promising in concept, hybrid energy systems involving tidal technologies remain in early demonstration phases and face significant engineering and regulatory hurdles before widespread deployment becomes feasible.

5) *Vertical Axis Turbines*: Vertical axis turbines (VATs) have re-emerged as a promising alternative to horizontal-axis designs due to their omnidirectional capabilities and mechanical simplicity. Unlike HATs, which require alignment with flow direction and often include complex yaw mechanisms, VATs can operate efficiently in reversing or turbulent tidal currents without reorientation. This feature makes them particularly attractive for deployment in estuarine and near-shore environments, where tidal currents may shift rapidly in both direction and intensity. VATs typically operate with a Darrieus or Savonius configuration, the former offering better energy conversion efficiency and the latter favored for startup torque and low-flow resilience [4]. Recent innovations in this cluster have focused on improving blade shape, mechanical durability, and startup performance, which historically limited the commercial viability of VATs. Although generally less efficient than HATs under ideal conditions, VATs offer superior reliability, reduced maintenance, and lower installation complexity, making them suitable for specific geographic and economic contexts. Ongoing work includes scale-model testing, structural optimization, and hybrid VAT-HAT system designs aimed at balancing efficiency with adaptability [4].

6) *Oscillating Hydrofoils*: Oscillating hydrofoils utilize the lift forces generated by water flow across a wing-like structure to drive a mechanical oscillation, which is then converted into rotational or linear motion for power generation. These systems differ fundamentally from traditional rotary turbines and offer several potential advantages, particularly in environments where conventional turbines may face ecological or navigational challenges. Oscillating hydrofoils typically operate at lower rotational speeds and with larger surface areas, which reduces the risk of marine life collisions and limits underwater noise generation. They are also visually less intrusive and may be more acceptable in protected or heavily regulated marine zones. However, the mechanical systems required to convert oscillatory motion into usable electrical energy are complex and often prone to biofouling, corrosion, and fatigue under high cycle loads [4]. Current research is directed toward improving mechanical linkages, minimizing energy losses at joints, and optimizing hydrofoil geometry for various flow regimes. While still in the experimental phase, oscillating hydrofoils are gaining traction as a viable technology for deployment in ecologically sensitive regions.

7) *Biofouling-Resistant Materials*: Biofouling—the accumulation of microorganisms, plants, algae, or animals on submerged surfaces—is a major operational challenge for tidal energy systems. This innovation cluster focuses on the development of materials and coatings that can resist biofouling, corrosion, and other forms of marine degradation [4]. Traditional antifouling methods, such as toxic coatings or frequent mechanical cleaning, have proven costly and environmentally

damaging. As an alternative, researchers are now exploring advanced composites, surface texturing, and low-energy-release coatings that physically or chemically deter organism settlement. These materials must also withstand high mechanical loads, saltwater exposure, and hydraulic stress while maintaining hydrodynamic efficiency. Some novel solutions include hydrophobic nanocoatings, electroactive surfaces, and biomimetic materials inspired by shark skin or mollusk shells. Implementation of such technologies can significantly extend turbine operational lifespan, reduce maintenance frequency, and increase net energy yield. However, long-term durability in real-world deployments and compatibility with structural requirements remain active areas of investigation.

8) *Floating and Modular Deployment Platforms*: Floating and modular deployment platforms are reshaping the physical infrastructure of tidal energy systems. Traditional seabed-anchored devices require complex and expensive marine construction, often limiting deployment to shallow waters with favorable geological conditions. Floating platforms eliminate the need for heavy foundations by using moored buoyant structures that support turbines at optimal depths. These platforms are particularly useful for deep-water sites or locations with soft or unstable seabeds. Modular designs enable easier transport, assembly, and maintenance, as components can be built off-site and rapidly deployed. They also facilitate scalability, allowing developers to begin with small arrays and expand incrementally [4]. From an engineering standpoint, challenges include maintaining platform stability under tidal and wave forces, optimizing tethering systems, and minimizing energy losses due to platform motion. Some floating designs also incorporate hybrid functions, such as supporting solar panels or monitoring equipment. As such, floating and modular platforms enhance not only flexibility and cost-efficiency but also the feasibility of deploying tidal energy in regions previously considered inaccessible.

9) *Adaptive Blade and Rotor Systems*: Adaptive blade and rotor systems represent a critical advancement in the pursuit of maximizing energy conversion efficiency under varying tidal flow conditions. Unlike fixed-pitch designs, adaptive systems enable real-time mechanical adjustments to the blade angle, chord, or curvature to optimize lift forces as water velocity fluctuates. One prominent strategy involves variable-pitch mechanisms that continuously adjust the angle of attack in response to real-time flow data, allowing turbines to maintain near-optimal performance across a broad range of velocities and tidal phases. These mechanisms not only improve energy capture but also reduce hydrodynamic stall events, which are common in fixed-pitch systems during low-flow or transitional periods between flood and ebb tides. In addition to variable-pitch strategies, several emerging designs feature morphing or shape-changing blades that dynamically adapt to hydrodynamic loading. These systems often rely on composite materials with embedded actuators or shape-memory alloys that deform the blade geometry in a controlled manner. Such deformations can reduce fatigue loads by smoothing transient flow impacts and can also minimize unsteady lift-drag fluctuations that contribute to structural vibration and wear. Rotors with passive adaptive characteristics—such as

flexible trailing edges or compliant mounts—have also been studied as cost-effective alternatives to actively controlled systems. The integration of adaptive rotor technology necessitates a sophisticated control system architecture, typically incorporating flow sensors, torque monitors, and real-time feedback algorithms. These systems must operate reliably in submerged environments, withstanding marine biofouling, corrosion, and pressure variability while maintaining actuator precision. Additionally, the mechanical complexity introduced by adaptive elements demands rigorous life cycle modeling, durability testing, and redundancy planning to avoid mission-critical failures. While adaptive systems offer substantial gains in energy yield and system responsiveness, they also introduce higher capital costs and operational risks, making them most viable in high-resource tidal zones where performance optimization can justify the added complexity.

10) Sensor-Integrated Maintenance and Diagnostics:

Sensor-integrated maintenance and diagnostic systems are transforming the reliability and longevity of tidal energy infrastructure by enabling continuous monitoring and predictive failure analysis. These systems embed a range of sensors within the turbine structure to measure critical operational variables such as stress, vibration, temperature, pressure, and acoustic signatures. For example, strain gauges placed along the turbine blades and shaft can detect cumulative fatigue, which is particularly important in high-cycle tidal environments where repeated loading may induce microcracking or structural degradation over time. Similarly, accelerometers and gyroscopes can track oscillations or imbalances caused by blade fouling, marine debris, or early-stage mechanical failure. Temperature and humidity sensors within electronic housings provide real-time feedback on internal thermal conditions, enabling early detection of electrical overheating or moisture ingress. This is particularly important for submerged power electronics, where failure of a waterproof seal can lead to rapid short-circuiting and irreparable damage. Acoustic sensors, both passive and active, are increasingly used to characterize cavitation events, gear misalignment, and even marine mammal activity in the vicinity, supporting both system health and environmental compliance monitoring. The deployment of these sensors enables condition-based maintenance (CBM) strategies, in which servicing is conducted only when specific indicators of wear or degradation are detected, rather than on a fixed schedule. This reduces downtime and operational costs, particularly for offshore deployments where maintenance mobilization is logistically complex and expensive. Furthermore, the volume of data generated by sensor networks facilitates the application of machine learning algorithms that can detect patterns and anomalies far earlier than traditional methods. For example, time-series analysis of vibration spectra can predict bearing failures before audible symptoms arise, allowing proactive replacement and extended equipment uptime. However, sensor systems must be carefully calibrated and engineered for the hostile marine environment. Issues such as biofouling on sensor lenses, electromagnetic interference from subsea cabling, and degradation of signal quality over long distances pose challenges for sustained performance. Power supply and data transmission must also be addressed

through autonomous energy harvesting or wireless data relays. Despite these hurdles, the proliferation of sensor-integrated diagnostics represents a paradigm shift in how tidal energy systems are maintained and evaluated, laying the foundation for autonomous, intelligent infrastructure capable of self-monitoring and self-healing behaviors in the long term.

11) Multi-Use and Ancillary Function Devices: Multi-use and ancillary function devices reflect an interdisciplinary shift in tidal energy system design, wherein power generation infrastructure is integrated with other marine, ecological, or industrial services to increase the value proposition of each installation. These platforms are often designed to fulfill secondary functions alongside electricity production, making them particularly attractive in regions with constrained infrastructure, high ecological value, or overlapping marine space uses. A prominent example of this cluster includes turbine housings and foundation structures that double as artificial reef habitats. These are engineered with complex geometries and eco-friendly materials that encourage colonization by marine flora and fauna, supporting biodiversity restoration in degraded ecosystems. The inclusion of textured surfaces and nutrient-retentive materials allows these installations to promote shellfish settlement, kelp attachment, and fish aggregation, thus transforming energy infrastructure into components of marine habitat rehabilitation. Other systems incorporate environmental monitoring functions, such as integrated sensors for measuring salinity, temperature, pH, turbidity, and dissolved oxygen. These sensors enable continuous oceanographic data collection for climate research, fishery management, and early-warning systems for harmful algal blooms or pollution events. Deploying such monitoring systems on energy infrastructure reduces redundancy and capital expenditure while enhancing environmental stewardship. Additional ancillary functionalities include the integration of desalination modules, which use tidal energy to power reverse osmosis processes for potable water production. These systems are particularly valuable in arid coastal regions and island nations, where freshwater scarcity is a chronic issue. Some designs have also explored support structures for aquaculture operations, enabling floating cages for fish or shellfish that benefit from the hydrodynamic flows around turbines. From an engineering perspective, the inclusion of multiple use cases introduces trade-offs in design constraints, structural loading, maintenance access, and environmental permitting. Shared structural components must be optimized for both mechanical performance and biological compatibility. Power distribution networks must be configured to accommodate variable load types, including non-critical services such as lighting, sensors, or communications equipment. Additionally, regulatory frameworks may lag behind these multi-functional designs, requiring cross-jurisdictional coordination among energy, fisheries, environmental, and maritime agencies. Despite these challenges, multi-use tidal energy platforms offer a compelling pathway toward maximizing the utility of marine space, reducing per-unit costs through shared capital investment, and enhancing the socio-ecological acceptability of marine renewable energy. By co-locating energy production with ecosystem services and local infrastructure needs, these systems embody a systems-level approach to

sustainable coastal development.

12) Further Innovation: Modern TECs span a range of architectures. Horizontal-axis (axial-flow) turbines, resembling underwater wind turbines, are the most mature. They offer high energy conversion efficiency when aligned with dominant tidal currents. However, they require active yaw control or bidirectional blades to handle reversing flows. Other variants, such as cross-flow Darrieus turbines, exhibit omnidirectional performance but suffer from lower efficiency. Oscillating hydrofoil systems convert lift forces into rotational motion, offering lower visual and environmental impact, but are mechanically more complex. Floating tidal kites, such as those developed by Minesto, allow deployment in deep water and low-velocity streams by using tethered wing systems to amplify water speed across the turbine. Paasch et al. highlight that design evolution is not limited to the turbine alone, but extends to support structures, anchoring systems, and maintenance access. They describe innovations in removable monopiles and telescoping foundations that reduce downtime and O&M costs, particularly in harsh Pacific coastal conditions. They also emphasize the role of hardware-in-the-loop simulators in validating new TEC concepts before field deployment [5]. Yet, despite rapid prototyping progress, field testing remains a major bottleneck. Rusvan et al. contribute a unique perspective through physical modeling of a two-way tidal reservoir using scaled laboratory prototypes. Their results suggest that flow aperture size and pond geometry significantly influence energy output, providing insights into optimizing hydraulic head and turbine flow rates [1]. This experimental approach helps bridge the gap between numerical simulation and real-world deployment, but also highlights the challenge of scaling fluid dynamics from lab to ocean. Lastly, the challenge of survivability in extreme marine environments persists. Corrosion, cavitation, marine growth, and mooring fatigue all impact long-term reliability. Some patents address these by introducing composite materials, active blade pitch control, and sensor-integrated maintenance systems, as reported in the BERTopic-based analysis by Pazhouhan et al. [4]. However, standardized qualification procedures—like those provided under the IEC 62600 series—are still in early adoption stages, and many prototypes fail to transition to commercial readiness. Tidal energy innovation is increasingly multi-dimensional, encompassing not just hydrodynamics but also materials science, structural engineering, and control system integration. But until these technologies mature through iterative sea trials and data sharing, their deployment will remain limited to pilot-scale projects.

D. Environmental Concerns

Environmental considerations are central to the engineering and deployment of tidal energy systems, particularly due to their location in ecologically sensitive marine and estuarine zones. As tidal turbines are often installed on or near the seabed, they interact directly with benthic habitats, sediment transport dynamics, and marine fauna. Multiple studies across the literature point to the need for integrated environmental impact assessments (EIAs) that combine hydrodynamic,

ecological, and regulatory perspectives. Paasch et al. present a detailed account of the regulatory hurdles faced by early-stage wave and tidal developers in the U.S. Pacific region, notably the Reedsport project by Ocean Power Technologies. They emphasize that environmental impact—particularly on fisheries and marine mammal migration patterns—is often the limiting factor in project licensing. To address these concerns, developers are increasingly required to conduct baseline assessments using tools like Acoustic Doppler Current Profilers (ADCPs) for flow characterization and underwater noise modeling to predict sound propagation from turbine operation [5]. Khojasteh et al. broaden this view, noting that recent research trends reflect a shift from resource-centric development to ecosystem-integrated planning. They identify emerging clusters focused on multi-criteria site selection frameworks, which factor in not only current velocity and tidal range but also biodiversity sensitivity, sediment erosion potential, and socio-political constraints [3]. In their patent landscape analysis, Pazhouhan et al. identify a rising volume of innovation focused on environmentally adaptive design. These include slow-rotating blade systems to reduce collision risk, blunt leading-edge geometries, and modular turbine housings designed to serve as artificial reefs, promoting marine biodiversity. While promising, these technologies remain largely at the prototyping or early demonstration phase and lack longitudinal performance data under real oceanic conditions [4]. Rusvan et al. contribute indirectly to environmental analysis through their modeling of tidal head reservoirs. Their work highlights how design changes in flow apertures and basin geometry can influence velocity gradients and flow acceleration, which in turn impact local sediment deposition and scour patterns [1]. These insights are valuable when considering deployments near deltas or estuarine systems where sediment transport is critical to habitat maintenance. One notable limitation in all reviewed studies is the short duration of post-deployment ecological monitoring. While temporary deployments offer early indicators of environmental compatibility, most EIAs are limited to 1–3 years of data, which is insufficient to capture chronic impacts such as species migration shifts or trophic cascade effects. Moreover, permitting agencies often operate on a precautionary principle, requiring mitigation plans even in the absence of observed impact, thereby increasing project cost and complexity. To address these challenges, several authors propose adaptive licensing frameworks, where small-scale demonstration projects collect impact data under controlled conditions, feeding into iterative environmental risk models. These frameworks are already being piloted in Europe under consortia like OES-Environmental, but are less prevalent in developing regions with fewer regulatory resources. While site-specific environmental modeling has matured significantly, there is a pressing need for long-term monitoring infrastructure, open access to environmental datasets, and standardization of ecological impact metrics. Engineering efforts must increasingly co-evolve with marine conservation principles to ensure sustainable tidal energy expansion.

E. Geographical Site Potential

The global potential for tidal energy is vast, with estimates suggesting up to 26,000 TWh/year of theoretical tidal resource availability, as noted in the large-scale bibliometric analysis by Khojasteh et al. [3]. However, the difference between theoretical and technically or economically extractable energy is substantial. This gap is influenced by a range of site-specific factors including tidal range (the vertical difference between high and low tides), current velocity (the horizontal flow speed of water), site bathymetry (the underwater topography that affects flow acceleration and turbulence), seabed composition (which determines anchoring feasibility and construction costs), and environmental or navigational constraints (such as shipping lanes or protected habitats). Several high-potential areas are repeatedly identified in the literature, including the Pentland Firth (United Kingdom), Cook Inlet (Alaska), Bay of Fundy (Canada), and Sunda Strait (Indonesia). These locations often feature narrow straits or coastal constrictions that enhance kinetic energy through topographic funneling of tidal flows. Rusvan et al. provide experimental context by demonstrating how a scaled laboratory model can be used to simulate tidal head differences and flow aperture dimensions, validating energy potential based on fluid geometry. Their results emphasize how channel shape and inlet constriction influence flow acceleration and energy density at a site [1]. The feasibility of tidal energy development is also linked to proximity to existing infrastructure and population density. Chen highlights that over half of the U.S. population resides within 50 miles of the coast, suggesting that tidal energy could serve as a highly localized and grid-proximate renewable source [6]. Yet many of the world's most energetic tidal sites lie in underdeveloped or remote regions such as eastern Indonesia, the Philippine archipelago, and parts of rural Alaska—areas where limited grid connectivity and high logistical costs hinder deployment. Thus, the spatial mismatch between energy potential and infrastructure readiness presents a key constraint. Patent landscape analysis by Pazhouhan et al. reveals that emerging device designs are increasingly tailored to specific geophysical contexts. For example, shallow estuarine turbines, floating deep-sea systems, and impounded lagoon-style technologies each correspond to distinct site profiles [4]. This suggests that as device diversity increases, techno-environmental matching may unlock new viable locations for tidal exploitation. Nonetheless, resource mapping remains inconsistent across regions. One significant limitation is the lack of high-resolution hydrodynamic datasets in the Global South, where oceanographic modeling tools and field measurements are scarce. While regions such as Europe and North America benefit from well-developed resources like NOAA tidal current atlases and UK Carbon Trust maps, many equatorial and island nations lack the wave, bathymetric, and current velocity data necessary to carry out detailed feasibility studies. This data deficiency severely limits early-stage site screening, especially in locations like Indonesia, which—despite its vast maritime territory of over 17,000 islands—has yet to fully develop its tidal resource base. A final consideration in assessing site viability is the

impact of turbine spacing and wake interactions in tidal arrays. Hydrodynamic interference between turbines placed too closely together can reduce efficiency through energy shadowing and turbulence. To address this, advanced modeling approaches such as Computational Fluid Dynamics (CFD) and Blade Element Momentum (BEM) theory are employed. CFD simulates fluid flow in three dimensions around turbine blades and channel structures, allowing for the optimization of array spacing and placement. BEM combines blade-specific lift and drag calculations with momentum theory to estimate turbine performance under various conditions. Together, these modeling tools are essential for maximizing extractable energy from even well-endowed sites by optimizing both individual turbine efficiency and system-wide energy yield. In summary, while tidal energy potential is globally distributed and technically abundant, the realization of this potential is constrained by data scarcity, logistical challenges, and mismatched infrastructure. Future progress will rely on the integration of satellite remote sensing, in-situ measurement campaigns, and model-driven site screening frameworks, paired with region-specific turbine designs.

F. Integration into Current Systems

Tidal energy's integration into existing and emerging energy systems is increasingly seen as a critical engineering challenge and opportunity, particularly for remote coastal communities, microgrids, and hybrid renewable configurations. The predictability of tidal cycles offers advantages in system planning and dispatch coordination, but its cyclic nature and non-continuous generation profile require integration with complementary technologies or storage systems to ensure uninterrupted supply. Klaus underscores the value of hybrid energy systems, demonstrating that a combination of Tidal Stream Energy (TSE), solar PV, and battery storage outperforms individual systems in terms of reliability and cost efficiency—especially in remote island contexts with high diesel fuel costs. His study shows that the longer operational hours of TSE (up to 20 hours/day) make it a better base-load complement than intermittent renewables like solar or wind. The hybrid model also enhances load matching, reducing battery oversizing and improving the levelized cost of electricity (LCOE) for consumers [2]. Khojasteh et al. report a growing research trend toward the co-location of marine renewables, including wave-tidal arrays, wind-tidal platforms, and multi-use offshore energy hubs. This approach enables shared use of subsea cabling, maintenance vessels, and grid infrastructure, thereby reducing capital expenditures [3]. However, such integration requires rigorous planning around electrical stability, harmonic distortion, and reactive power compensation, particularly in weak-grid or stand-alone systems. Environmental compatibility also plays a role in integration. For example, in Rusvan et al.'s prototype experiments on two-way tidal reservoirs, the setup mimics the functionality of a mini pumped-storage system, storing water during high tides and releasing it through turbines at low tide to smooth output over a 6-hour tidal cycle. While the laboratory model is small-scale, it demonstrates how tidal impoundment can

serve as both an energy generator and a dispatchable storage mechanism [1]. A particularly important use case for tidal energy is in powering desalination plants, where predictable but cyclic power availability is acceptable. According to Chen, marine energy can be used to operate reverse osmosis systems, particularly in arid coastal regions where freshwater access is limited and energy costs are high. The modular and off-grid nature of such installations aligns well with tidal generation profiles [6]. From a systems engineering perspective, the integration of tidal energy into broader grids requires attention to grid code compliance, especially for frequency stability and fault ride-through capabilities. In isolated systems, tidal generators must incorporate power electronics (inverters, voltage regulators) and SCADA systems for active control and islanding detection. These are areas still under development, as many TECs remain focused on hydrodynamic efficiency rather than electrical compatibility and grid harmonization. Lastly, Pazhouhan et al. suggest that future integration pathways may involve ocean energy clustering, where patents show a rise in multi-output devices capable of both power generation and ancillary services (e.g., marine navigation lighting, environmental sensing, or aquaculture aeration). These multi-functional deployments could increase return on investment and open up niche markets where standard utilities are absent [4]. In conclusion, while tidal energy's predictable output is an asset in system design, technical, electrical, and operational integration challenges persist. Future progress depends on cross-disciplinary solutions, involving electrical engineers, marine planners, and utility regulators to fully embed tidal power into local and regional energy systems.

G. Tidal Energy Policy

Tidal energy is increasingly recognized as a strategic contributor to global energy-transition efforts, particularly in the context of decarbonization, energy diversification, and climate resilience. Its unique characteristics—predictable generation, low carbon emissions, and high energy density—make it a valuable addition to renewable-energy portfolios aimed at achieving net-zero targets. Multiple papers in the literature underscore the importance of aligning tidal-energy development with national and international policy frameworks to overcome its current limitations in cost, scale, and visibility. Khojasateh et al. emphasize that tidal energy, though technologically promising, remains marginal in global deployment—contributing only 1.6 TWh globally in 2020 compared to over 23,000 TWh of total electricity demand. Despite this, they point to a recent shift in research focus from purely technical assessments to systems-level thinking, such as integrating tidal energy into coastal-management plans and developing multi-criteria decision-making frameworks for site prioritization [3]. These shifts align with broader sustainability goals outlined in policies such as the European Green Deal and the United Nations' Sustainable Development Goals (SDGs). Klaus identifies a major barrier to policy alignment as the lack of market parity and reliable financial modeling for tidal-stream energy (TSE) [2]. Traditional financial-appraisal methods often overlook the long-term social and

environmental value of tidal systems. Klaus therefore argues for the adoption of social discount rates (SDRs)—discount rates that reflect society's preference for present versus future welfare, typically lower than private discount rates—to ensure that cost-benefit analyses capture the enduring public value of zero-carbon energy. In tandem, Klaus calls for explicit carbon externality pricing (e.g., carbon taxes or emissions-trading schemes) so that the societal cost of greenhouse-gas emissions is internalized in project economics. Together, appropriately calibrated SDRs and carbon-pricing mechanisms can rebalance investment appraisals, allowing tidal-energy projects to compete more favorably in government-funding schemes and carbon-credit markets [2]. Chen supports this perspective by positioning tidal energy as a complementary technology that enhances grid resilience owing to its predictability and consistency—attributes that are especially important in high-renewables-penetration scenarios where solar and wind dominate but are variable [6]. In this context, tidal systems can function as load-following base generation, supporting grid-frequency stability and reducing reliance on fossil-based peaking plants. Several papers also address the role of standardization and regulation in unlocking policy support. For example, the IEC 62600 series—developed to standardize marine-energy performance, safety, and testing protocols—is beginning to gain traction among developers and regulators. Khojasateh et al. note that such standards are critical in reducing perceived investment risk and enabling eligibility for green-finance instruments such as green bonds and renewable-energy subsidies [3]. Pazhouhan et al. contribute a macro-level view by showing that patent activity is heavily concentrated in policy-forward countries like China, the UK, and Canada, where government-led initiatives—such as the UK's Contracts for Difference scheme or Canada's Ocean Supercluster—have provided targeted funding and regulatory support for pilot and pre-commercial tidal-energy projects [4]. This suggests that policy alignment is a direct catalyst for innovation and commercialization. Nevertheless, policy fragmentation remains a major hurdle. In many jurisdictions, marine-energy permitting is handled by multiple overlapping agencies, often creating extended timelines and inconsistent standards. Paasch et al. describe how the U.S. regulatory environment for wave and tidal projects has been hampered by conflicting jurisdiction between federal and state bodies, particularly in the areas of environmental-impact reviews and offshore rights-of-way [5]. To address this, some researchers call for adaptive regulatory frameworks that allow early-stage projects to deploy under experimental licenses with flexible monitoring requirements, thus enabling both innovation and environmental protection. These models have begun to appear in places like Scotland and Portugal but are not yet globally adopted. In summary, tidal energy's role in the clean-energy transition is increasingly acknowledged in academic and policy circles, but its advancement depends heavily on supportive governance, transparent carbon pricing, socially appropriate discounting of future benefits, financial incentives, and standardized evaluation criteria. Aligning technological development with policy innovation will be crucial to unlocking the full potential of this predictable, carbon-free energy source.

H. Tidal Energy Commercialization

Despite notable progress in tidal-energy research and demonstration, moving from successful prototypes to commercially viable deployments remains a core challenge. This well-known “valley of death” in technology development is repeatedly documented in the literature, with a consensus forming around several key barriers: high capital intensity, installation complexity, maintenance risk, and the absence of standardized operational data. Klaus underscores that tidal-stream energy (TSE) is still an emerging sector, with global installed capacity lingering at roughly 32 MW as of 2020—negligible when compared with wind and solar power, which together exceed 1,000 GW. He notes that early expectations for rapid commercialization were often premature: developers scaled too quickly or underestimated the realities of turbine installation, retrieval, and servicing in high-flow marine environments [2]. Khojasheh et al. show that many high-profile failures in tidal energy stemmed from unproven durability under fatigue loading, biofouling effects, and underdeveloped maintenance logistics. For example, turbine blades and moorings exposed to marine growth and variable flow conditions have exhibited faster-than-expected degradation, while maintenance windows in fast-moving tidal straits can be limited to less than 30 minutes per cycle [3]. These constraints demand engineered modularity, such as quick-release mooring systems and surface-retractable turbines—features that remain rare in commercial designs. Pazhouhan et al., in their patent-landscape study, reveal vibrant innovation in areas such as biofouling-resistant coatings, advanced blade materials, and dynamic-load management. Yet they stress that many prototype deployments are privately funded, and the resulting performance data are treated as proprietary [4]. Because these hard-won operational insights are seldom shared, cumulative learning slows and replication or scaling by other developers becomes more difficult. Paasch et al. provide an important historical view, showing that even promising deployments—such as Ocean Power Technologies’ Oregon site—were plagued by licensing delays, inter-agency coordination problems, and site-specific cost overruns [5]. These cases demonstrate that scaling is not only a technical hurdle but also a regulatory and project-management challenge, often underestimated when moving from single-device pilots to multi-turbine arrays. From a hydrodynamic standpoint, scaling introduces additional complexities. Rusvan et al. model a two-way tidal reservoir and show that small changes in flow aperture or basin dimensions can significantly alter velocity profiles and energy output. Extrapolating to full-scale arrays suggests that inter-device interference, turbulence, and sediment-scouring effects must be thoroughly simulated and tested before deployment [1]. Many current array layouts still rely on trial-and-error spacing, leading to wake losses and performance degradation. Financing remains another significant constraint. Klaus observes that most tidal projects are front-loaded with long payback periods and uncertain returns, making them unattractive to traditional investors without de-risking mechanisms such as grants, loan guarantees, or feed-in tariffs [2]. In contrast to wind and solar, supported by mature supply chains and

global component markets, tidal energy still lacks standardized platforms—raising costs and slowing adoption. A promising pathway cited by several authors is the modularization of deployment systems, where turbines are designed for rapid assembly, transport, and retrieval—paralleling offshore wind’s evolution. Floating foundations that remove the need for heavy seabed anchoring could further enable deployment in deeper or sediment-prone sites. In sum, tidal-energy technology has reached impressive experimental milestones, yet commercial scaling remains hindered by intertwined structural, logistical, and economic barriers. Addressing these challenges will require coordinated investment in field testing, data standardization, supply-chain development, and streamlined regulation, alongside cross-sector collaboration to move the sector from prototype to power plant.

III. RESEARCH LIMITATIONS

A. Disparity in Tidal Models

While tidal energy is a promising field, there are still some significant limitations to the current technology and research. The limitations of the research presented here are largely related to the lack of standardization and commercialization of tidal energy technology, as touched on previously. The lack of commercialization results in a broad and dispersed collection of research, with each new paper using a different model or design. While it could be helpful to have information on a broad range of designs for tidal power plants, the lack of focus or standardized models has ultimately slowed the implementation process and impact of this technology as a whole.

Khojasteh et al. paints an excellent picture of the scattered research landscape surrounding tidal energy. In their review of wave and tidal energy over the last twenty years, Khojasteh et al. found a broad range of topics, with limited focus on one specific type of tidal energy system and no frontrunner for tidal energy systems becoming apparent. Of their identified key words, only 10% related to tidal energy, demonstrating how tidal energy research has been underrepresented thus far. Khojasteh et al. raises the important point that, due to the lack of commercial maturity of tidal energy technology, research has generally shifted towards exploring novel approaches and designs that would optimize energy efficiency and conversion [3]. This lack of a specific focus in tidal energy research results in a limitation in the amount of information given for one specific design or implementation of this technology.

An example of this lack of focus on one particular tidal energy model can be seen in comparing Aliffathur Rusvan et al. to Chen and the different technologies they cover. Rusvan et al. covered a case study on a two-way tidal barrage system designed for a waterway in Indonesia. This system used a typical tidal barrage setup consisting of a concrete barrier to block off a tidal basin from the ocean and a number of flow holes with turbines inside that rotated with the rise and fall of the tides [1]. Chen, on the other hand, states that the industry has not stayed focused on a tidal barrage system, and describes the newer tidal stream energy systems as the most optimistic way forward for this technology [6]. Tidal stream

devices are individual turbines that could be connected in an array and placed in the ocean to operate in a similar way as wind turbines. Chen notes that there is not one particular tidal energy technology that has emerged as a frontrunner, and thus there still remains a high initial cost in this technology due to the lack of supply chain and bulk manufacturing. Although it is clear that there is yet to be a consensus or frontrunner in the search for an optimal tidal energy design, there are paths forward for how this research limitation could be overcome. Paasch states that the first step towards a clear design focus is increased in-situ testing [5]. Current designs must be tested both in a laboratory setting and, most importantly, in an ocean setting to accurately portray their performances and move forward towards a more focused research landscape.

B. Analytical Focus of Current Research

Closely related to the statement by Paasch regarding the need for more field testing, another significant impact of the lack of standardization of tidal energy technology in the current tidal research field is a focus on analytical or theoretical research, rather than to scale experimentation or data collection and analysis. As a frontrunner in this technology has yet to be identified, most research around this tidal energy as a renewable energy source has been theoretical or analytical. This can be seen clearly in the 1:10 laboratory scale model that Aliffathur Rusvan et al. use in their experiments. As there are a minimal number of tidal systems operating across the globe, most research and data around these systems and possible designs have been lab based or entirely mathematical [1]. While research in this field is vital to its development, the lack of deployment of physical models up to this point has limited the progress.

Another example of this can be seen in the patent landscape analysis done by Pazhouhan et al. Pazhouhan et al. found that, while most patents associated with wave energy are found to be related to maximizing the efficiency of existing wave conversion devices, the largest cluster of tidal energy patents are associated with novel designs for tidal energy systems. The majority of research and development is focused on theoretical designs, rather than testing or collecting data on current tidal power plants [4].

The lack of real world data and reliance on theoretical models is a massive limitation for tidal energy research. While models can be accurate to a degree, they will never approach the accuracy of field testing and experimentation. Klaus notes in his economic analysis of tidal energy research that the next step for this research is to analyze data from industrial prototyping and to move on from the demonstration phase and towards commercialization and implementation. Klaus suggests that many researchers see tidal energy as having completed the demonstration and ideation phases and ready for data acquisition and design optimization [2].

The limitations of the current research surrounding tidal energy are closely tied to the stage of development of this technology. Although humans have been using tidal kinetic energy for thousands of years, the large-scale implementation of tidal energy into the power grid is still a relatively new

and unexplored sector of renewable energy. As a result, the research surrounding this field is broad and covers a variety of design ideas, rather than focusing on one specific, ideal design. Additionally, there is a lack of data acquired from existing tidal systems. Instead, most of the research is centered around models and theoretical analysis. Both of these are limitations in the current research that will not be overcome without further development in the field, including the implementation of new full-scale tidal energy systems and extensive in-situ testing of these designs.

C. Intermittent Energy Source

The nature of tidal energy technology is that it relies on a naturally occurring phenomenon, over which humans have no control. While tides have the benefit of predictability, there is also a limitation tied to the intermittent intervals at which they occur. High tide might not happen at a time of peak energy consumption and the pattern of the tides overall is unlikely to align with human grid behavior. This could lead to many challenges similar to those that solar energy faces in regards to the curtailment of energy during peak production and the need for storage methods.

Chen touches on this limitation of tidal energy when describing the differences between tidal and wave energy. Chen writes that wave energy is entirely reliable as it is available constantly and year-round, while solar and wind are unpredictable [6]. Tidal energy is somewhere between the two, as it is more predictable than solar or wind power, but is not as constantly available as wave energy. Its predictability could aid in the planning of its usage as tides can be projected years in advance, however it is not the continuous switch of energy that wave energy provides. While a slight limitation, this is still a factor to consider when discussing tidal energy as it could require more planning and funding in its integration.

Rusvan et al. mention a similar note on the cyclical and intermittent quality of tidal energy production. In their report, however, they also note the differences in tidal heights with each season. It was touched on in a prior section that tides can generate for a maximum of twenty hours in a day, with an average of ten hours of generation. The neap and spring tides can alter this generation by up to four hours [1]. This intermittency greatly limits the amount of energy tides are able to provide, as they are bound by celestial timing.

The benefit of tidal predictability years in advance still remains as a massive attraction for investors and researchers as many of common renewable energy sources, like solar and wind, are limited in their predictability. This predictability does not negate the fact that tidal energy is still an intermittent source, limited to a maximum production time of ten discontinuous hours a day. The discontinuity and limit to energy production are both factors that must be taken into consideration as energy storage and hours of operation are planned for future tidal power plants. This limitation could be overcome through proper planning and the implementation of storage techniques to more evenly distribute the energy produced by a tidal power plant.

D. Corrosion and Maintenance Cost

A final limitation with tidal energy that must be considered when conducting further research or implementing this technology is the cost associated with maintenance and the corrosion tidal technology may face. With any device that is placed in the ocean, biofouling and corrosion from the saltwater will be a limitation. While cost of installation due to a lack of supply chain has been discussed, there will be additional costs related to corrosion resistant coatings, the research associated with this, and the maintenance costs of the technology after a period of time.

There are a few authors who touched on the limiting factor of the cost of corrosion on tidal power plants. Pazhouhan et al. mention corrosion resistance and ocean energy infrastructure as technological challenges still facing this field [4]. Chen additionally writes of the damage that powerful saltwater wave movement could do on expensive equipment [6]. There is missing, however, an in depth review of the lifespan of tidal designs and the cost that maintenance would add to the total cost of the technology. As protective coatings against saltwater corrosion can be expensive and the damage from the corrosion irreparable, this is a significant limitation to tidal energy and a missing factor from tidal energy research thus far.

This limitation could be solved through further research and discussion on the materials that make up tidal systems and their interactions with saltwater. In addition to this, in-situ testing of tidal models to determine the lifespan and potential cost of coatings or repairs from saltwater corrosion would further this gap or limitation in the field of tidal power technology.

IV. UNANSWERED QUESTIONS IN THE FIELD

A. Resource Characterization and Predictability

Despite many improvements in oceanographic modeling, these models often cannot aid in the predictability of the energy yield of each potential build site. Paasch et al. covers this topic in some detail in their paper. They highlight this issue in the context of the Pacific coast, noting that Oregon has some of the highest energy flux in the world during winter months, but the wave energy varies drastically over short distances due to the local topography [5]. This makes this region an unsatisfactory location to build upon.

This variability is especially true for wave energy, which remains inconsistent due to the deep complexity that is ocean dynamics. It is increasingly difficult to accomplish when also accounting for the atmospheric, hydrodynamic and climate models for better energy forecasting. Compounding all of these different variables into one predictable model is an ongoing challenge which has yet to be solved.

Moreover, this model could change in the future, with climate change being so unpredictable in current times. Changes in sea level, oceanographic patterns, weather frequency, and other factors could significantly alter both the model required to predict the effectiveness of a build site, and the actual effectiveness of the site itself. The lack of long-term data and climate projections tailored to marine energy resource predictions make it difficult to determine site viability, and the

lack of data makes it difficult to start modeling a system which could acceptably model the complex relationships between tidal energy and the things which affect it.

B. Technological Development and Optimization

As of now, there is still no generally agreed upon consensus for the standard design of WEC technology. The lack of a standard design or accepted “best” design overall means that the cost of producing this technology is expensive, as there is no standardized way to manufacture parts or build sites. Optimization challenges extend from the manufacturing and building portion of WEC and TEC all the way to the actual performance of the sites, not to mention the case by case variability previously stated above.

Marine environments are varied, and harsh, meaning that many materials can degrade quite quickly. No consensus has been made on how to deal with and mitigate these conditions, whether it be through the use of different materials, or the coating of materials with other chemicals to protect them. On top of this, WEC and TEC is still quite a small portion of research, and thus it is difficult to advance in the front. Khojasteh et al. covers this topic in their paper, where they compiled the total number of Ocean energy based papers in the last 20 years, from 2003-2023, and found that only 8174 papers were published on the matter. Moreover, about 50% of the papers published were exclusively from China, the U.S., and the U.K., showing just how under researched oceanographic energy currently is [3].

On top of this, there is the question of Hybridization as a means of optimization. Pazhouhan et al. and Klaus both advocate for multi-use offshore platforms as the future of ocean energy systems. They argue that the use of wind energy, wave converters, and subsea storage could provide grid flexibility and lower marginal costs, with the main caveat being that these systems are still purely theoretical and idealistic, and are not yet being studied or developed [4] [2].

C. Environmental Impact of Tidal Energy Sites

The ecological impact of Tidal and wave energy conversion sites holds a great importance when discussing TEC and WEC. With much of the technology relying on the use of turbines or rotors, these methods can harm local marine animals, as well as interrupt migratory paths and sediment transport on the ocean floor. These areas of study have limited empirical data, making it difficult to come to a conclusion about the viability of TEC and WEC in terms of its impact on the environment and ecosystem which surrounds it.

Moreover, there is a lack of consensus on how one even goes about quantifying these impacts effectively and legitimately. Paasch et al. discuss the gap in our understanding of the ecological and environment impacts, noting that most environmental impact assessment models rely on models from offshore wind [5]. Very few studies directly mention; let alone measure the disturbance to marine life from long term use of WEC or TEC. Rusvan et al. also touches on this, raising concern over the unknown cumulative impacts of large-scale tidal deployments, as these effects have yet to be studied [1].

As such, we have yet to verify what (if any) impact these technologies may have on sediment transport, water quality, and biodiversity on a large scale.

D. Economic Viability

Paasch et al. state that the cost can vary dramatically depending on which technology is being deployed, on top of changes from each site, and assumptions about maintenance costs [5]. Chen makes the conclusion that tidal energy has a more predictable resource profile, but suffers from high uncertainty in performance and cost [6]. Many of the issues of economic viability stem from the lack of development and optimization, which was touched upon above. The standardization of WEC and TEC is not yet in place, making its production and construction significantly more expensive than that of other energy conversion systems.

Currently, the levelized cost of energy (LCOE) for wave and tidal projects still remains significantly higher than that of more mature sectors of renewable energy, with estimates around \$200-\$400 per MWh, while offwind shore energy systems average \$40-\$60 per MWh, and solar averaging \$20-\$50 per MWh. This alarming increase in cost disincentives further construction of sites utilizing TEC and WEC due to its expensive nature.

Chen notes that wave energy devices specifically struggle with high costs per installed kW due to the mechanical complexity, survivability issues, and overall low energy density per unit. Tidal energy, while more predictable than wave energy, often requires subsea installations, which can dramatically increase the price of construction, and decrease efficiency due to an increased LCOE [6].

This can be attributed to the lack of full-scale developments which can be used to analyze these costs, on top of the constant exposure to harsh marine conditions, which can cause the deterioration of materials rapidly, hiking up costs of TEC and WEC systems. Reducing the risk of investing in TEC and WEC technology remains a challenge which is difficult to tackle due to this uncertainty. Both Pazhouhan et al. and Klaus note that the lack of insurance and financing models regarding this marine sector inhibits its growth, as companies are wary to invest in such unpredictable, untested technologies [4] [2].

V. POSSIBLE EXTENSIONS IN THE FIELD

A. Hybrid Systems

Some possible extensions in WEC and TEC technologies can include the use of hybrid systems, taking advantage of multiple different renewable energy sources. There has been some talk of eventual oceanic platforms which would be built in order to take advantage of solar, wind, and wave energy converters. These are great ideas in theory, but have not yet been realized. Some of the reasons as to why this has not yet been done includes engineering complexity and high up front costs. While this technology has been individually established, it has not yet been engineered to be combined and produced commercially for platforms which could utilize various forms of renewable energy. Hopefully, as the technology advances, we will see more development in these hybrid technologies.

B. Energy Storage Integration

Another possible extension in the field of WEC and TEC technologies includes the integration and utilization of energy storage systems to provide a fully off-grid energy solution of island microgrids, as well as other use cases. Embedding energy storage systems like lithium-ion batteries, supercapacitors, or flywheels, would allow for enhanced grid smoothing, reduced curtailment, and black-start capabilities. WEC technologies specifically are fairly variable, as the amount of energy which can be harvested from waves depends on many factors, making it a somewhat unreliable form of energy production. The use of energy storage systems would allow for excess energy harvested to be stored for later use, when wave energy may die down, allowing the energy to be supplemented from storage. Moreover, this would enable the restarting of grid sections during outages, enhancing overall grid robustness and resilience.

C. AI Base Control Systems

A prominent possible extension in the field which could greatly optimize the operation of WEC and TEC technologies would be the use of predictive algorithms and machine learning to optimize the operation of WEC and TEC technologies based on set trends and patterns. This use case of AI and algorithms could maximize the total energy harvested, while reducing the mechanical stress on the components of the systems, as well as account for predictive maintenance based on what may need to be repaired, detected by subtle changes in vibrations or output to flag any issues preemptively. This extension has not yet been fully realized, as not many models for predicting tides and waves, let alone extremely accurate and complex ones. Waves and tides have many external factors which complicate algorithmic predictions, however, if these challenges are overcome, the use of predictive algorithms could greatly improve the efficiency and cost effectiveness of WEC and TEC technologies as a whole, boosting their viability.

VI. CONCLUSION

Tidal energy as a commercialized energy source is still a relatively new technology in the field of renewable energy. Throughout the research presented, all six papers had a tone of optimism for the current contributions to the field and the possible future extensions. Although some limitations, gaps in the research, and unanswered questions exist, tidal energy technology as a whole is a promising addition to the diverse set of renewable energy sources that will define the future energy grid.

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